

letAlcook — Full Technical Documentation

Project Vision

letAlcook is an AI engineering coordination platform designed to automate project planning, Jira task management, sprint coordination, and deployment guidance.

System Goals

- Transform software ideas into structured engineering roadmaps
 - Automatically generate technical tasks
 - Assign tasks intelligently using AI
 - Sync tasks with Jira
 - Track dependencies and blockers
 - Improve software team coordination
-

High-Level Architecture

```
Frontend (Next.js)
  ↓
FastAPI Backend
  ↓
Project Service
Task Generation Engine
Skill Matching Engine
Sprint Planning Engine
Jira Integration Service
AI Orchestrator
  ↓
PostgreSQL + Redis
  ↓
OpenAI API + Jira API
```

Backend Architecture

```
backend/
├── app/
│   ├── api/
│   ├── core/
│   ├── db/
│   ├── models/
│   ├── schemas/
│   ├── services/
│   ├── ai/
│   ├── workers/
│   └── utils/
```

Frontend Architecture

```
frontend/
├── app/
├── components/
├── hooks/
├── services/
├── lib/
└── styles/
```

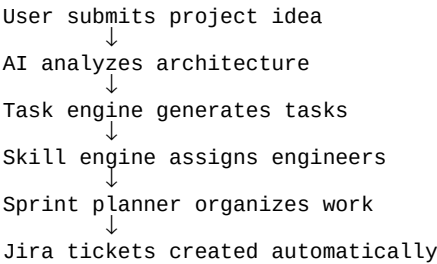
Core Services

1. Project Service 2. Task Generation Engine 3. Skill Matching Engine 4. Sprint Planning Engine 5. Jira Integration Service 6. AI Orchestrator 7. Deployment Automation Engine

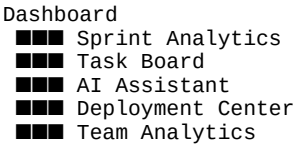
Database Design

users developers projects milestones tasks task_assignments sprints ai_logs project_memory jira_sync_logs

AI Workflow



UI Architecture



Recommended Development Roadmap

Week 1-2: - Backend foundation - Authentication - Database design Week 3: - AI roadmap generation Week 4: - Frontend dashboard Week 5: - Jira integration Week 6: - Skill matching engine Week 7: - Sprint planning engine Week 8: - Deployment automation

Future Scaling

Future improvements: - Kubernetes orchestration - GitHub repository analysis - Autonomous sprint optimization - AI engineering analytics - Infrastructure automation
